

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project R26-SE-009 | Agent 4 | Malith P. Bandara (IT22249166) | Reporting period 23.03.2026 – 25.08.2026

1. PROJECT AND STUDENT DETAILS

Project ID	R26-SE-009
Research Topic	Self-Optimized LLM Multi-Agent System for Software Engineering
Supervisor	Prof. Nuwan Kodagoda
Co-Supervisor	Mr. Eishan Weerasinghe
Student Name	Malith P. Bandara
Student Registration No.	IT22249166
Individual Component	Agent 4 — Deployment Engineer (AI DevOps / Deployment) and the DevOps user interface
Reporting Period	23 March 2026 to 25 August 2026
Phase Covered	Requirement, implementation and integration phase, representing approximately 65% of the planned project work.

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

This reporting period covers the development of Agent 4, the deployment and DevOps component of the multi-agent system, together with the DevOps-facing user interface. The period began by fixing the deployment outputs the project requires — container assets, compose files, workflow pipelines, cloud configuration and release evidence — and by designing the pipeline checkpoints, the artefact set, the readiness score and the gate that blocks a release while QA issues remain open. The first implementation followed, together with two iterations of the deployment interface covering the release intake conversation, workflow change inspection, timeline replay, diff preview and artefact presentation, and contributions to the shared studio screens. An end-to-end review of that first path identified the checks missing before a real release, which shaped the rebuild: environment and database validation, credential and secret handling, a deployment event and state model, a provider abstraction and monitoring views. Alongside the rebuild, the repository gained its CI quality and security gates and a functional integration dashboard with connection diagnostics.

3. DATE-WISE RP WORK (44 ENTRIES)

MARCH 2026

4 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
1	23.03.2026	Defined the scope of Agent 4 as the AI deployment engineer responsible for packaging, readiness and release after QA approval.
2	25.03.2026	Reviewed the expected deployment outputs: container assets, compose files, workflow pipelines, cloud configuration and release evidence.

NO.	DATE	RP WORK CARRIED OUT
3	27.03.2026	Designed the workflow that moves approved source code through to deployment artefacts and a readiness report.
4	31.03.2026	Identified the pipeline checkpoints for source validation, service discovery, security review, packaging, deployment and documentation.

APRIL 2026

9 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
5	02.04.2026	Defined the artefact set required from the component: compose files, workflow files, task definitions, summaries and evidence bundles.
6	06.04.2026	Designed readiness scoring and the deployment gate so unresolved QA issues block release preparation.
7	09.04.2026	Planned the reporting structure for operational evidence such as services, images, logs and API validation results.
8	13.04.2026	Implemented the deployment assistant interface and the conversation area used for release intake.
9	16.04.2026	Implemented workflow change viewing and filtering so modified pipeline files are inspected before a release.
10	20.04.2026	Implemented timeline replay for deployment runs together with the diff preview section.
11	23.04.2026	Implemented the deploy start and status endpoints with run tracking.
12	27.04.2026	Implemented artefact generation for compose files, workflow files and the deployment summary.
13	30.04.2026	Reworked the deployment interface, removing superseded sections and improving how artefacts are presented.

MAY 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
14	05.05.2026	Completed version 2 of the deployment workspace and reviewed it with the group.
15	08.05.2026	Built the design-selector screens and the preset browsing flow in the shared workspace.
16	12.05.2026	Built the deployment timeline and the progress states shown while a release is running.
17	15.05.2026	Added the generated-page preview and the pane used to inspect produced files.
18	19.05.2026	Expanded the deployment views to cover services, workflows and container images.
19	22.05.2026	Added the readiness score panel and the evidence checklist to the DevOps views.
20	26.05.2026	Completed version 2.1 of the deployment workspace following the internal review.
21	29.05.2026	Reviewed the deployment path end to end and listed the checks missing before a real release could be attempted.

JUNE 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
22	02.06.2026	Compared container, CI and cloud provider options and selected the target deployment routes.
23	05.06.2026	Designed the environment validation carried out before a deployment is allowed to start.
24	09.06.2026	Designed credential and secret handling so no deployment secret reaches the repository or the logs.
25	12.06.2026	Planned the CI quality and security gates to be added to the repository workflows.
26	16.06.2026	Specified the deployment event and state model needed to track a running release.
27	19.06.2026	Designed the provider abstraction so more than one deployment target can be supported.
28	23.06.2026	Specified the monitoring views covering pipeline, repository, security and evidence.
29	26.06.2026	Reviewed the rebuilt deployment design with the group and closed the open points.

JULY 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
30	01.07.2026	Added the SonarCloud quality-analysis workflow and configured project analysis for the repository.
31	06.07.2026	Added the Snyk dependency and infrastructure-as-code security workflow.
32	09.07.2026	Documented the quality and security connection steps and the environment configuration they require.
33	13.07.2026	Built the functional integration access dashboard and the integration status service.
34	16.07.2026	Enabled the integration controls in the DevOps Center and exposed connection diagnostics.
35	21.07.2026	Added the repository CLI integration card and cleaned generated files out of version control.
36	24.07.2026	Fixed the CI tests for the quality and security workflows and corrected the container validator environment.
37	29.07.2026	Fixed the front-end build and upgraded the dependencies flagged by the security advisories.

AUGUST 2026

7 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
38	03.08.2026	Rebuilt the deployment package on the new module structure.
39	06.08.2026	Implemented environment validation and the database connectivity check run before a deployment.
40	11.08.2026	Implemented credential and secret handling for deployment operations.
41	14.08.2026	Implemented deployment event and state tracking for running releases.
42	18.08.2026	Implemented the release generator for container, workflow and cloud configuration output.
43	21.08.2026	Refreshed the package and flow contracts shared with the QA stage and strengthened validation.

NO.	DATE	RP WORK CARRIED OUT
44	25.08.2026	Implemented provider authentication and the deployment API used for provider-backed releases.

4. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

- 1. Complete provider-backed deployment through Vercel and the AWS route, including release generation and rollback handling.
- 2. Finish deployment monitoring — pipeline, repository, security and evidence views — in the DevOps Center.
- 3. Strengthen readiness scoring and secret handling so releases are gated on verifiable QA acceptance.
- 4. Extend the generated deployment artefacts into more complete service-specific container and cloud configurations.
- 5. Tighten the live handoff from Agent 3 to Agent 4 so deployment starts only after a strong QA pass signal.

5. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

SUPERVISOR'S COMMENTS

Malith P. Bandara STUDENT — IT22249166 Date: _____	Prof. Nuwan Kodagoda SUPERVISOR Date: _____	Mr. Eishan Weerasinghe CO-SUPERVISOR Date: _____
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